

An Adiabatic Quantum Algorithm for the Traveling Salesman Problem

Pierre-Emmanuel WULFMAN

Graduate School of Mathematics



26 May 2025

Outline

- 1 Introduction
- 2 Traveling Salesman Problem (TSP)
- 3 Adiabatic Algorithm for TSP
- 4 Complexity

Why Adiabatic Quantum Computing?

- Alternative model to the quantum circuit model
- Promising for optimization and NP-complete problems
- Grounded in physical principles: quantum adiabatic theorem
- D-Wave and other quantum annealers are practical implementations

This presentation draws heavily from the work of (Kieu 2019)¹

¹kieu_2019.

Quantum Adiabatic Theorem

Theorem (Simplified)

Given a time-dependent Hamiltonian $H(t)$, if a quantum system starts in the ground state of $H(0)$ and $H(t)$ evolves slowly enough, then the system remains in the ground state of $H(t)$ for all t .

- Adiabatic Quantum Computing (AQC) leverages this theorem to solve optimization problems.

Model Overview

Key Ideas

- Start with H_I with easy-to-prepare ground state.
- Encode the problem in H_P whose ground state encodes the solution.
- Create a time-dependent Hamiltonian
$$H(t) = (1 - t/T)H_I + (t/T)H_P.$$
- Evolve the system from H_I to H_P over time T .
- Measure the final state to obtain the solution.

A solution is the minimal of a cost function $C(x)$, and the ground state of H_P corresponds to the state of minimal energy.

Pros and Cons of Adiabatic Quantum Computing

Pros

- Naturally suited for optimization problems
- Robust to certain types of noise and decoherence
- Can leverage existing quantum annealing hardware
- Conceptually simple: follows physical intuition

Cons

- Requires slow evolution (long computation times)
- Gap size can limit efficiency (may be exponentially small)
- Not universal for all quantum algorithms
- Hardware limitations and control precision are challenging

Traveling Salesman Problem (TSP)

Problem Statement

Given a set of cities and distances between them, find the shortest possible route that visits each city exactly once and returns to the origin city.



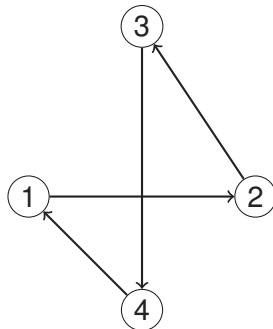
Formal Definition

- Input: A set of N cities and a distance matrix $D = [d_{ij}]$.
- Objective: Find a permutation π of $\{1, 2, \dots, N\}$ that minimizes:

$$\sum_{i=1}^N d_{\pi(i), \pi(i+1)}$$

- With the constraint: $\pi(N+1) = \pi(1)$ (return to start).

Visual Example



Example of a TSP tour visiting 4 cities

Computational Complexity

- The number of possible tours grows factorially $O(N!)$
- No known polynomial-time algorithm exists
- Best known algorithms (quantum) in $O^*(1.728^n)$.²
- Possible candidate for quantum speedup

²Andris Ambainis et al. (n.d.). “Quantum Speedups for Exponential-Time Dynamic Programming Algorithms”. In: *Proceedings of the 2019 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 1783–1793. DOI: 10.1137/1.9781611975482.107. eprint: <https://epubs.siam.org/doi/pdf/10.1137/1.9781611975482.107>. URL: <https://epubs.siam.org/doi/abs/10.1137/1.9781611975482.107>.



Preamble

The algorithm uses creation-annihilation operators to create H_I and H_P .

Path between cites

$$[l_{ij}, l_{i'j'}^\dagger] = \delta_{ii'} \delta_{jj'}$$

number of traversals

$$|n\rangle_{l_{ij}}$$

Relations

$$l_{ij}^\dagger |n\rangle_{l_{ij}} = \sqrt{n+1} |n+1\rangle_{l_{ij}}$$

$$l_{ij} |n\rangle_{l_{ij}} = \sqrt{n} |n-1\rangle_{l_{ij}} \quad (n \geq 1)$$

$$l_{ij} |0\rangle_{l_{ij}} = 0$$

$$\hat{n}_{ij} = l_{ij}^\dagger l_{ij}$$

Concept

Maximally untangled state

$$|\{\theta\}\rangle_I = \bigotimes_{ij} |\theta_{ij}\rangle$$

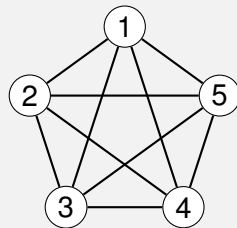
where

$$|\theta_{ij}\rangle = e^{-|\theta_{ij}|^2/2} \sum_{n=0}^{\infty} \frac{(\theta_{ij})^n}{\sqrt{n!}} |n\rangle_{l_{ij}}$$

We have,

$$l_{ij}|\{\theta\}\rangle_I = \theta_{ij}|\{\theta\}\rangle_I$$

Maximally Connected Graph



Every city is connected to every other city.

Filtering

We want to select for a tour that is

- *complete* : every city is visited at least once
- *connected* : no disjoint subgraphs

Two new creation-annihilation pairs :

"markers"

$$[m_i, m_j^\dagger] = \delta_{ij}$$

$$m_i |0\rangle_{m_i} = 0$$

Marks the cities that have been visited.

rmk : the set of $\{m\}, \{h\}, \{l\}$ commutes.

"hooks"

$$[h_i, h_j^\dagger] = \delta_{ij}$$

$$h_i |0\rangle_{h_i} = 0$$

Selects loops in the tour.

Filters

Departure from city 1

$$F = \sum_{j=2}^N m_j^\dagger h_j^\dagger \hat{n}_{j1}$$

Traveling from city i to city j

$$L = \sum_{\substack{i,j=1 \\ i \neq j}}^N m_j^\dagger h_j^\dagger \hat{n}_{ji} h_i$$

Returning to city 1

$$E = \sum_{j=2}^N \hat{n}_{1j} h_j$$

Putting it all together

Combination operators

$$Q = \underbrace{\left(\prod_{i=2}^N m_i \right)}_{\text{remove incomplete}} \underbrace{E(F)^{N-2} L}_{\text{All tours from 1 visit city once}}$$

Eigenvector

$$|\{n\}\rangle_{\{l\}} \otimes |\{p\}\rangle_{\{m\}} \otimes |\{q\}\rangle_{\{h\}}$$

Rmk : After filtering, not all tours are valid, but those that are not will not corresponds to a ground state of H_P .

Putting it all together (2)

Eigenvalues of Q

$$Q|\{n\}\rangle_{\{l\}} \otimes |\{0\}\rangle_{\{m\}} \otimes |\{0\}\rangle_{\{h\}} = \left(\prod_{n_{ij} \in \text{tour}} n_{ij} \right) |\{n\}\rangle_{\{l\}} \otimes |\{0\}\rangle_{\{m\}} \otimes |\{0\}\rangle_{\{h\}}$$

- It is easy to see that a tour which use a path several times will have a higher energy than one that uses each path once.
- It follows that $(Q^\dagger - 1)(Q - 1)$ annihilates all the complete tours, where each path is traversed exactly once.

Hamiltonian

Initial Hamiltonian

$$H_I = \sum_{i,j=1}^N (l_{ij}^\dagger - \theta_{ij}^*)(l_{ij} - \theta_{ij}) + \sum_{i=1}^N m_i^\dagger m_i + \sum_{i=1}^N h_i^\dagger h_i$$

This Hamiltonian has a simple ground state

$$|g_I\rangle = |\{\theta\}\rangle_{\{I\}} \otimes |0\rangle_{\{m\}} \otimes |0\rangle_{\{h\}}$$

Final Hamiltonian

Final Hamiltonian

$$H_P = s \left(\sum_{i=1}^N h_i^\dagger h_i + \sum_{i=1}^N m_i^\dagger m_i \right) + s(Q^\dagger - 1)(Q - 1) + \sum_{i,j=1}^N d_{ij} \hat{n}_{ij}$$

With

$$s = \frac{1}{2} \sum_{i,j=1}^N d_{ij}$$

- The first part ensures that the $|g_P\rangle$ ends in $|0\rangle_{\{m\}} \otimes |0\rangle_{\{h\}}$,
- The second promotes connected complete tours with no double traversals.
- The third part remove links that are not part of the tour.



Final ground state

Final ground state

$$|g_P\rangle = |\text{solution}; \{n\}_{\text{others}} = 0\rangle_{\{l\}} \otimes |0\rangle_{\{h\}} \otimes |0\rangle_{\{m\}}$$

Complexity Analysis

Resource complexity

$O(N^2)$ links, $O(N)$ markers, and $O(N)$ hooks.

Time complexity

Using the *Schrödinger* equation :

$$i\frac{\partial}{\partial t}|\psi(t)\rangle = H(t)|\psi(t)\rangle$$

One can find the min time T to get $|g_P\rangle$ with a target probability

$$T_{\perp} \sim O\left(\frac{1}{\Delta_I E_P} \int_0^1 g(\tau) d\tau\right),$$

Lower bound given in [Tien D. Kieu \(2006\). Quantum Adiabatic Computation and the Travelling Salesman Problem](#) [arXiv:](#)



Explanation

Energy spread of the $|g_I\rangle$ in H_P

$$\Delta_I E_P = \sqrt{\langle g_I | H_P^2 | g_I \rangle - \langle g_I | H_P | g_I \rangle^2}$$

proportional to the energy gap between the ground state and the first excited state of H_P . As $T_\perp = T_\perp(s, \{\theta\})$, for $\theta_{ij} = \theta$,

Approximation for large N and small θ

$$\Delta_I E_P \sim O\left(s\sqrt{(N-1)!}|\theta|^N\right)$$

Exponential reduction in the time complexity at the expense energy for preparation.

Exact Time complexity is still an open problem.

Conclusion

- We gave an example of AQC algorithm design
- Complexity analysis shows potential for exponential speedup over classical algorithms.
- Other promising result for NP-complete problems (e.g., *3-SAT*, *Knapsack*, etc.)
- Still less popular than circuit model, but has practical implementations (e.g., D-Wave systems).

Thank You!

Questions?

References I



Ambainis, Andris et al. (n.d.). “Quantum Speedups for Exponential-Time Dynamic Programming Algorithms”. In: *Proceedings of the 2019 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 1783–1793. DOI: 10.1137/1.9781611975482.107. eprint: <https://epubs.siam.org/doi/pdf/10.1137/1.9781611975482.107>. URL: <https://epubs.siam.org/doi/abs/10.1137/1.9781611975482.107>.



Kieu, Tien D. (2006). *Quantum Adiabatic Computation and the Travelling Salesman Problem*. arXiv: quant-ph/0601151 [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/0601151>.

References II



Kieu, Tien D. (Feb. 2019). "The travelling salesman problem and adiabatic quantum computation: an algorithm". In: *Quantum Information Processing* 18.3. ISSN: 1573-1332. DOI: 10.1007/s11128-019-2206-9. URL: <http://dx.doi.org/10.1007/s11128-019-2206-9>.